BÁO CÁO THỰC HÀNH LAB 04

GUI Programming with Swing

Student Information

- Name: Le Kim Phu
- Student ID: 20235808
- Course: Object-Oriented Programming
- Lab: Lab 04 – GUI Programming with Swing

1. Objectives of the Lab

In this lab, I learned how to:

- Create GUI applications using Java Swing
- Understand the differences between AWT and Swing
- Use layout managers to organize GUI components
- Handle events using ActionListener
- Convert part of the AIMS project from CLI to GUI

2. AWTAccumulator and SwingAccumulator

2.1 AWTAccumulator

The AWTAccumulator application uses AWT components such as:

- Frame
- Label
- TextField

The program allows the user to:

1. Enter a number
2. Press Enter
3. Add the number to the total sum
4. Display the updated sum

The event is handled using ActionListener and actionPerformed().

2.2 SwingAccumulator

The SwingAccumulator application performs the same task as AWTAccumulator but uses Swing components:

- JFrame
- JLabel
- JTextField

Unlike AWT, Swing components are added to the content pane of the JFrame.

3. Comparison Between AWT and Swing

Feature	AWT	Swing
Components	Heavyweight	Lightweight

UI Design	Uses native OS UI	Java-based customizable UI
Window	Frame	JFrame
Text Label	Label	JLabel
Text Field	TextField	JTextField
Flexibility	Less flexible	More powerful and flexible

Swing provides a more modern and customizable graphical interface than AWT.

4. Layout Managers

4.1 GridLayout

GridLayout arranges components in rows and columns equally. This layout manager divides the container into a grid of equal-sized cells. Each component occupies exactly one cell, and all cells have the same dimensions.

Example:

- NumberGrid keypad layout

4.2 BorderLayout

BorderLayout divides the window into five regions:

- NORTH
- SOUTH
- EAST
- WEST
- CENTER

Used in:

- StoreManagerScreen

5. NumberGrid Application

The NumberGrid application contains:

- Digit buttons
- DEL button
- C button
- Display text field

Functions

- Digit buttons append numbers to the display
- DEL removes the last digit
- C clears the entire display
- The buttons share a common ActionListener

6. GUI for AIMS Project

In this lab, the AIMS project starts converting from a console application to a GUI application. The application is divided into:

5. Store Manager

6. Customer (implemented in future labs)

7. StoreManagerScreen

The StoreManagerScreen displays all media items in the store. This is the main interface for store administrators to manage inventory.

Components

- Menu bar
- Header
- Media grid

Features

- View store
- Add Book
- Add CD
- Add DVD

The CENTER section uses GridLayout to display media items.

8. MediaStore

MediaStore extends JPanel and represents one media item in the GUI. Each MediaStore component is responsible for displaying information about a single media item.

It displays:

- Media title
- Media cost
- Buttons

If the media implements Playable, the Play button is shown. This allows users to preview playable media items like CDs and DVDs.

9. Event Handling

ActionListener is used for:

- Menu item events
- Button events
- Play button actions

The actionPerformed() method handles user interactions. This method is called whenever a user performs an action that triggers an ActionEvent, such as clicking a button or selecting a menu item.

10. Inheritance in GUI Screens

To reduce duplicated code, inheritance can be used. By creating a parent class that contains common functionality, we can create multiple child classes that inherit from it, each with specific behavior for different media types.

Parent Class

- AddItemToStoreScreen

Child Classes

- AddBookToStoreScreen
- AddCompactDiscToStoreScreen
- AddDigitalVideoDiscToStoreScreen

This improves reusability and maintainability. By sharing common code in the parent class, we eliminate redundancy and make future changes easier to implement across all child classes.

11. Conclusion

This lab helped me understand:

- Java Swing programming
- GUI component organization
- Event handling
- Layout managers
- Applying OOP concepts in GUI applications
- Converting CLI applications into GUI systems

The lab also prepared the foundation for future GUI development in the AIMS project. The knowledge of Swing components, layout managers, and event handling provides a solid base for implementing more complex features such as customer interfaces, shopping carts, and transaction management in subsequent labs.